



Engineering science

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Synthesis of Energetics

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» Chapter 1 — Sufficiency

1.1 Context and Motivation

Global climate targets are under pressure. The 6th IPCC report (Working Group 3) confirms with high confidence that the world is behind on addressing climate change. Most 2°C scenarios (two thirds of them) rely on negative carbon technologies such as Direct Air Capture (DAC) or Bioenergy with Carbon Capture and Storage (BECCS), yet these technologies have been studied for over 50 years without becoming economically viable.

The remaining global carbon budgets are tight. Updated 2023 estimates put the budget at approximately 260 GtCO₂ for a 50% chance of limiting warming to 1.5°C, and around 950 GtCO₂ for a 67% chance of staying below 2°C. For context, proven fossil fuel reserves embed some 3 500 GtCO₂ of potential emissions — far beyond any safe budget.

1.2 The Sufficiency–Efficiency–Renewables Framework

The presentation applies the *negaWatt* approach, built on three complementary levers formalized through the Kaya identity:

- **Energy sufficiency** — reducing the level of a need through behavioural and societal change (e.g. smaller dwellings, less car use, dietary shifts). Introduced in IPCC AR6 WG3 as a policy category in its own right.
- **Energy efficiency** — reducing the energy required to deliver a given service through better technology (heat pumps, insulation, electric vehicles).
- **Renewable energy** — decarbonizing the remaining energy supply.

A key conceptual distinction is made between sufficiency and efficiency: efficiency improvements can be cancelled out by growing consumption (the rebound effect). German residential data illustrate this clearly — gains in energy intensity per m² have been largely offset by rising living space per capita.

1.3 Sectoral Demand Scenarios

The *negaWatt* Belgium scenario covers agriculture, residential, tertiary, mobility, freight, industry, non-energy-demand (NED) industry, and aviation bunkers.

› 1.3.1 Buildings

Residential space heating (65 TWh in 2019) is projected to fall by 71% to 19 TWh in 2050, driven by:

- Temperature setpoint reduction: 21 °C → 19 °C (–12%)
- Building envelope renovation at 3%/year from 2030 (–28%)
- Heating technology switch (18% → 82% heat pumps and district heating) (–20%)
- Improved system efficiency, average COP 2.5 → 3.5 (–10%)

Combined with the tertiary sector, total building final energy consumption falls by 48% between 2019 and 2050, with natural gas and liquid fuels being largely replaced by ambient heat and electricity.

› 1.3.2 Transport

Individual car mobility (56 TWh in 2019) is reduced by 93% to just 4 TWh by 2050 through a combination of sufficiency and efficiency measures:

- Distance reduction (−10%) via teleworking and urban densification
- Modal shift: car share 80% → 57% (−14%)
- Car occupancy: 1.25 → 2 persons per vehicle (−18%)
- Near-full fleet electrification, 99.5% by 2050 (−31%)
- Vehicle energy efficiency gains: 0.22 → 0.13 kWh/km (−19%)

The number of cars on Belgian roads would fall from 6 million to 2 million. Domestic transport final energy consumption declines by 83% overall.

› 1.3.3 Industry

Key industrial transformations include:

- **Steel:** −15% volume reduction; fuel switch from coke to hydrogen (60% direct reduction) and electric arc furnaces (40%); energy intensity 3 950 → 2 300 MWh/kt.
- **Cement:** −25% volume (more timber construction); energy intensity 904 → 650 MWh/kt; fuel switch coal → biomass.
- **Chemicals:** −20% demand; process electrification and fuel switching to renewables.

Total industrial final energy consumption falls by 34%.

› 1.3.4 Overall Final Energy Consumption

Belgium's total final energy consumption decreases from **490 TWh in 2019** to **246 TWh in 2050**, a reduction of **50%**, broken down as: buildings −49%, transport −76%, industry −34%.

1.4 Energy System Modelling: PyPSA-Belgium

The supply side is modelled using the open-source PyPSA-Eur framework adapted for Belgium. The model features:

- One node per country, covering five countries to represent cross-border interconnections
- Optimization runs for 2030, 2040, and 2050 to map the full transition pathway
- Hourly time resolution to capture renewable variability and flexibility requirements
- Full sector coupling: electricity, gas, hydrogen, heating, and industry

Sufficiency demand assumptions from the negaWatt Belgium scenario are fed directly as inputs to the supply-side optimization.

1.5 Two Scenarios Compared

	Reference (Ref)	Sufficiency (nW BE)
Emission target (Belgium, 2050)	−95%	−95%
Target (5 countries, 2050)	Net zero	Net zero
Demand	Current trends	Sufficient demand
Carbon Capture and Storage (CCS)	Allowed	Not allowed
Carbon Capture and Utilization (CCU)	Allowed	Allowed
Direct Air Capture (DAC)	Allowed	Not allowed
Building renovation rate	1%/year	3%/year
Industry	Continued trends	Increased circularity

1.6 Key Results

› 1.6.1 Cost-Effectiveness

The negaWatt Belgium scenario reduces total average annual system costs (CAPEX + OPEX) by **33% compared to the reference scenario** and by 19% compared to 2019 levels. Savings arise from reduced fuel import costs, avoided infrastructure investment, and lower operating costs due to a smaller overall energy system.

› 1.6.2 Energy Independence

The sufficiency scenario doubles Belgium's share of local energy production (from approximately 50% to 62%) and reduces energy imports by a factor of four. This significantly improves strategic autonomy, especially in combination with circular economy policies that limit dependence on critical raw materials.

› 1.6.3 Renewable Deployment

Sufficiency substantially reduces the renewable capacity required:

- **Solar PV:** 48 GW (reference) vs. 48 GW (sufficiency), but annual installation rate drops from 3 000 MW/year to 1 600 MW/year.
- **Onshore wind:** 14 GW (reference) vs. 7.4 GW (sufficiency); annual additions 440 MW vs. 160 MW.
- **Offshore wind:** 8 GW in both scenarios.

› 1.6.4 Hydrogen

Hydrogen demand in 2050 reaches 82 TWh in the sufficiency scenario, which is **43% less** than the 144 TWh needed in the reference case. Hydrogen is reserved for hard-to-decarbonize uses: industry, long-distance transport, synthetic fuel production, and non-energy applications. Most supply (84%) is imported; local electrolysis provides 13 TWh.

› 1.6.5 Cumulative Emissions

The sufficiency scenario achieves the –95% emission target for Belgium by 2050 without relying on CCS or DAC, using instead biomass-based carbon cycles, land use, and forestry (LULUCF) sinks. The reference scenario reaches the same 2050 target but relies heavily on large-scale carbon removal, resulting in significantly higher cumulative emissions over the transition period.

1.7 Sensitivity Analyses

› 1.7.1 Additional Offshore Wind

Expanding offshore capacity by an additional 16 GW (to 24 GW total by 2050, consistent with the Esbjerg and Ostend declarations) reduces onshore wind needs by 35% and increases power-to-gas capacity by 66%. Electricity and hydrogen imports also decrease. Core messages on cost-effectiveness and feasibility remain unchanged.

› 1.7.2 Nuclear Energy Costs

At the default cost assumption of 8 594 EUR/kW, new nuclear is not selected by the optimizer. Nuclear begins to appear in the solution only at approximately 4 500 EUR/kW — a dramatic cost reduction only achievable with small modular reactors (SMRs) under optimistic assumptions. At that price point, around 10 GW of nuclear capacity would be deployed, substituting for some onshore wind, solar PV, and reducing transmission needs.

1.8 Key Takeaways

1. **Cheapest scenario:** The negaWatt Belgium pathway is *more* cost-effective than a technology-only reference scenario, reducing system costs by 33%.
2. **Strategic independence:** Consuming less energy directly increases Belgium's energy autonomy and reduces exposure to geopolitical energy risks.
3. **Realistic scenario:** The sufficiency scenario avoids CCS and DAC, technologies that face major industrial-scale deployment challenges. Renewable deployment remains ambitious but stays well below estimated technical potentials.
4. **Fair scenario:** The scenario prioritizes collective solutions benefiting the most vulnerable citizens, treats affordable energy access as a fundamental right, and frees up global resources for the Global South by reducing Belgium's overall footprint.

The full model and data are openly available at:

- <https://github.com/IntSusEnergySystems/PyPSA-Belgium>
- <https://www.negawatt.be/scenario/>

» Chapter 2 — Energy market

Energy Markets — Executive Synthesis

Based on irex Consulting's lecture for UCLouvain, February 2026

2.1 Market Structure & Key Actors

The energy market involves several distinct layers of actors operating across the **energy supply chain** and the **physical infrastructure**:

- **Producers** — Generate electricity or gas (e.g. ENGIE, Luminus, TotalEnergies, Aspiravi). They dispatch power plants and inject energy into the grid.
- **Energy Markets** — Centralised trading platforms (e.g. EEX, EPEX SPOT, Nord Pool, ICE) where energy is bought and sold.
- **Balance Responsible Parties (BRPs)** — Entities responsible for balancing their portfolio of supply and demand on the market.
- **Suppliers** — Commercial intermediaries (e.g. ENGIE, Luminus, Lampiris, Eneco, Mega) who purchase energy and provide it to end consumers.
- **Transmission System Operator (TSO)** — Manages high-voltage transport of energy across the network.
- **Distribution Grid Operator (DGO)** — Distributes energy at lower voltage to end consumers.
- **Consumers** — Residential and industrial end-users.
- **Regulator** — Oversees market fairness and compliance (e.g. CREG, VREG, CWaPE, Brugel in Belgium).

2.2 Production Mix — Belgium

Historical Trends (2010–2025)

Belgium's electricity production mix has undergone significant transformation:

- **Nuclear** has declined from 55% (2010) to $\approx 34\%$ (2025), but remains a dominant source.
- **Coal** has been phased out between 2010 and 2020 (from 7% to 0%).
- **Wind and Solar** have grown rapidly, collectively reaching $\approx 32\%$ in 2025.
- **Gas (CCGT)** plays a flexible backup role ($\approx 19\%$ in 2025).

2026 Policy Update: Nuclear Is Back

The new Belgian government has repealed the nuclear phase-out law. Key developments:

- Long-term operation (LTO) extensions of **Doel 4** and **Tihange 3** are being negotiated until 2045/2055.
- A **50/50 public-private joint venture** between the Belgian State and ENGIE will manage the 2 GW LTO capacity.
- Nuclear is re-confirmed as a long-term pillar of the Belgian energy mix alongside renewables.

2.3 Green Electricity Labels

Three mechanisms certify electricity as “green” in Belgium:

1. **Green Certificates (GsC)** — Tax/subsidy mechanism for green producers; an obligation for suppliers, but not a consumer-facing label.
2. **Guarantees of Origin (GoO)** — Tracks renewable energy production; allows suppliers to market supplied energy as green. Currently at yearly granularity, with initiatives exploring finer time resolution.
3. **Power Purchase Agreements (PPAs)** — Direct physical purchase of electricity from producer to consumer.

2.4 Gas Supply — Belgium

2018 vs. 2025 Comparison

	2018	2025
Total Import	343 TWh	428 TWh
Domestic Consumption	187 TWh	150 TWh
Total Export	156 TWh	278 TWh
<i>Main import sources (2025):</i>		
Norway	48%	38%
LNG ships	8%	34%
Netherlands	32%	(net exporter)
UK	4%	13%
France	—	15%
<i>Main export destinations (2025):</i>		
Germany	—	50%
Netherlands	—	14%
France	44%	—

The key shift is the **East/West balance disruption** caused by the Russia–Ukraine conflict: Belgium switched from exporting to France to becoming a major transit hub exporting to Germany and the Netherlands, while LNG imports surged dramatically.

2.5 Energy Price Formation

The Merit Order

Electricity prices are set by the **marginal technology** needed to satisfy demand at any given hour. Technologies are dispatched in order of increasing marginal cost:

Wind/Solar → Nuclear → CCGT → Oil

- **Normal case:** CCGT sets the marginal price (≈ 60 €/MWh).
- **High RES production:** Renewables or nuclear set a low marginal price.
- **Low RES & high demand:** Oil or peaking units set a very high price (> 130 €/MWh).

The Clean Spark Spread (CSS)

When gas-fired plants (CCGT) are marginal, the electricity price is driven by gas and CO₂ costs:

$$\text{CSS} = P_{\text{elec}} - 2 \times P_{\text{gas}} - 0.411 \times P_{\text{CO}_2}$$

As of early 2026, with gas at ≈ 26 €/MWh and EUAs at ≈ 80 €/tCO₂, a CCGT requires power prices above **85 €/MWh** to be profitable. The Belgian forward price (CAL 27 BASE) stood at 76.27 €/MWh, implying CCGTs are *out of the money* (CSS ≈ -9 €/MWh).

Price Evolution (2021–2025)

- **2021:** Rising prices driven by post-COVID industrial restart, low wind, hot summer, and low French nuclear availability.

- **2022:** All-time high prices following Russia's invasion of Ukraine and supply chain disruption. Spot prices peaked at ≈ 450 €/MWh.
- **2023–2025:** Significant price correction due to supply chain reorganisation, good weather, and high gas storage levels. Prices remain above pre-COVID levels.

2.6 Supply Contracts

Energy suppliers offer contracts spanning a spectrum from fully fixed to fully variable:

	Fixed Price	Variable Price
Mechanism	Single €/MWh price locked at signing	Price varies each hour with spot market
Price driver	Futures market	Day-Ahead SPOT
Hedging strategy	Supplier buys in advance	Supplier buys on SPOT DA
Customer risk	Low	High
Supplier risk	High	Low

Index and *Click* products sit between the two extremes.

2.7 Supplier Differentiation

Since electricity and gas are commodities (identical products), suppliers cannot differentiate on product quality. Competitive levers are:

- **Price** — With razor-thin gross margins ($\approx 5\%$), cost efficiency is paramount.
- **Customer service** — Operational excellence in billing, switching, and support.
- **Marketing** — Green image, local identity, brand trust.
- **Extra services** — eMobility, flexibility management, apps, maintenance.

2.8 Energy Market Evolutions

Cannibalization Effect

As the share of renewables increases, a structural challenge emerges: the **cannibalization effect**.

When large volumes of renewables produce simultaneously, the supply curve shifts right, depressing prices at exactly the hours of peak renewable output. This reduces the *captured price* of each additional unit of renewable capacity:

$$\text{Captured price} = \frac{\sum_h \text{Prod}_h \times \text{Price}_h}{\sum_h \text{Prod}_h}, \quad \text{Captured ratio} = \frac{\text{Captured price}}{\text{Price}_h}$$

Between 2021 and 2024, **solar captured value dropped by 30–40%** across major EU grids (Spain, France, Germany, Greece).

Negative Prices

Negative electricity prices are becoming frequent. In Belgium:

- **519 hours** cleared at negative prices in 2025, representing $\approx 6\%$ of all hours.
- This is a more than threefold increase compared to 2024 (≈ 400 hours).

2.9 Flexibility: The Key Enabler of the Energy Transition

The grid must remain balanced at all times (Energy IN = Energy OUT). With rising shares of intermittent renewables, **flexibility** becomes critical.

$$\text{Flexibility} = \text{Adjusting the demand or supply curve}$$

Flexibility Strategies

1. **Local optimisation** — Control peak power (kW), reduce grid fees, manage capacity tariffs.
2. **Market optimisation** — Day-Ahead (DA) and Intraday (ID) arbitrage, imbalance (IMB) optimisation.
3. **Reserve services** — Frequency regulation (aFRR, R1/R2/R3), voltage control for the TSO.

The value of flexibility ranges from ≈ 5 k€/MW/year (peak management) to ≈ 300 k€/MW/year (primary frequency reserve R1/R2).

Battery Energy Storage Systems (BESS)

BESS is the most scalable flexibility asset:

- Global investment in BESS reached **\$65.5 billion** in 2025 (source: IEA).
- In 2024, a 1–3 h BESS could earn up to **0.5 M€/MW** through combined DA/ID and ancillary market revenues.
- At current ancillary market prices, BESS projects can be profitable in **less than 5 years**.
- However, a **massive development pipeline** across EU markets (BE, DE, NL, FR, UK) risks saturating ancillary markets.

Other Flexibility Sources

- **Electric Mobility** — Smart (V1G) and bidirectional (V2G) charging.
- **Industrial demand response** — Electrolysers, process interruptions.
- **Residential** — Heat pumps, boilers, home batteries, solar curtailment.

All of these are enabled by **dynamic pricing tariffs**.

2.10 New Market Actors

The energy transition has given rise to new roles in the value chain:

- **Prosumers** — Consumers who also produce energy (e.g. households with solar panels and batteries), actively participating in both supply and demand.
- **Aggregators** — Entities that pool flexibility from many small assets and offer it to the BRP or TSO. They sign framework agreements with BRPs and flexibility contracts with prosumers.
- **Energy Service Companies (ESCOs)** — Provide energy solutions, optimisation insight, and turnkey services to industrial and commercial clients.

» Chapter 3 — Wind turbines

3.1 Summary: Ventis – Wind Farm of Gouy-Lez-Piétons

» 3.1.1 Company Overview

Ventis S.A. is a Belgian wind energy developer founded in the early 2000s by brothers Benoît and Pierre MAT, headquartered in Tournai. The company manages wind energy projects entirely in-house – from development and permitting through construction and long-term operation – operating with a local and sustainable philosophy in Wallonia and France. Its internal team covers all key disciplines: development, construction, operation, administration, and communication.

Between 2022 and 2025, Ventis installed 25% of all new wind capacity commissioned in Wallonia, recently becoming the first independent wind energy producer in the region. The company currently operates approximately 90 wind turbines across Belgium and France, representing a total installed capacity of around 250 MW, of which nearly 200 MW corresponds to its own ownership share.

In Belgium, Ventis operates wind farms across the provinces of Hainaut, Brabant wallon, and Liège, with sites in operation such as Dour–Quiévrain–Hensies (9 turbines), Tournai-Antoing-Brunehaut (10 turbines), La Louvière (5 turbines), and Quévy-Mons (7 turbines), among others. Several projects are also authorised or under development, including repowering operations and new greenfield sites. In France, Ventis operates farms in Longueville-sur-Aube (Aube, 11 turbines) and Gorges-Gonfreville (Manche, 9 turbines), with additional projects under development in Pas-de-Calais and Aisne.

» 3.1.2 The Gouy-Lez-Piétons Project

Permitting History

The Gouy-Lez-Piétons wind farm illustrates the complexity of the Belgian permitting process for wind energy. Ventis submitted its initial permit application in March 2014. The first decision, issued in August 2014 by the delegated and technical officers, was a refusal based on an unfavourable opinion from Belgocontrol (the Belgian air navigation service provider). Ventis appealed, obtained a conditional opinion from Belgocontrol, and was granted the permit on appeal by Minister Carlo Di Antonio in January 2015.

However, the municipality of Pont-à-Celles lodged a further appeal in March 2015, citing high landscape impact. The Council of State annulled the permit in March 2017 on the grounds that alternative locations had not been sufficiently examined. Ventis resubmitted the permit application in May 2019, providing a supplemental impact study. The permit was re-granted by the Minister in July 2019, followed by yet another appeal from the municipality in September 2019, this time contesting the examination of alternatives. Finally, in November 2022, the Council of State cleared the permit of all appeals, confirming that no alternative locations were available. From first application to final clearance, the permitting process spanned nearly nine years.

Technical Characteristics

The wind farm comprises 8 turbines arranged across agricultural land near Gouy-Lez-Piétons, connected by underground MV cables and internal access roads. The main technical characteristics are as follows:

- **Wind turbine model:** Nordex N131
- **Nominal power per turbine:** 3.6 MW
- **Total installed capacity:** 28.8 MW
- **Total height:** 150 m
- **Rotor diameter:** 131 m

- **Grid connection:** DSO ORES, 10.8 kV, 25 MW injection capacity
- **Annual energy production:** ≈ 57 GWh
- **Capacity factor:** $\approx 26\%$
- **Households supplied:** $\approx 14\,400$ (yearly average consumption)

Ownership Structure

Six of the eight turbines are owned by the project company GOUYNRG, while the remaining two are owned by ENGIE, which was a competing developer during the development phase. The GOUYNRG shares are distributed as follows: 70% held by Ventis, 15% by Greenplug (a development partner), and 15% by local public and citizen participation – reflecting Ventis’s commitment to community involvement and local acceptability.

› 3.1.3 Project Development

Wind Turbine Sizing

The dimensions of the wind turbines are constrained by a combination of biological and air traffic factors. The minimum ground clearance (distance between the blade tip at its lowest point and the ground) is determined by biological constraints, particularly the presence of bat species requiring a safety margin above the tree canopy. The maximum blade tip height is set by air traffic constraints enforced by Belgocontrol. Together, these two values define the admissible rotor diameter (their difference) and, consequently, the tower height. The wind resource assessment further informs the optimal rotor size, since power extracted from the wind scales with the swept area and the cube of wind speed:

$$P_{\text{wind}} = \frac{1}{2} \rho A v^3 \quad [\text{W}]$$

where ρ is air density (approximately 1.2 kg/m^3 at 15°C), A is the rotor swept area, and v is wind speed.

Turbine Layout and Wake Effects

The number and spatial distribution of turbines is governed by the available grid injection capacity (25 MW with ORES), land acquisition constraints, local setback distances from housing, natural environments, and infrastructure, and wake effect losses. Wake effects occur when a downstream turbine operates in the turbulent, lower-energy airflow created by an upstream turbine. The prevailing wind rose at the site (predominantly south-southwest to west-southwest) directly influences the optimal turbine spacing and alignment to minimise these losses.

A constraint map overlaying setback distances from housing zones, motorways, high-voltage lines, hertzian beams, forests, and Natura 2000 areas was produced during the development phase to define the feasible turbine positions.

› 3.1.4 Turbine Technology

Turbine Selection Process

Three candidate turbine models were evaluated: the Vestas V126 (3.45 MW), the Nordex N131 (3.60 MW), and the Siemens Gamesa SG-132 (3.47 MW). Selection criteria included permit height limits (total height and ground clearance), rotor size, nominal power, noise output, site access feasibility, wind resource studies, and site assessment reviews.

A key finding during the site assessment was that turbine E3 (and to a lesser extent E6) faced a 15% energy curtailment constraint due to the proximity of a small woodland, reducing the net annual energy production of the SG-132 option to only 49 130 MWh/year – lower than the N131’s 57 688 MWh/year. Combined with a competitive price-to-performance ratio and no turbulence-related power limitation, the Nordex N131 was selected.

Generator Architectures

Two main electrical architectures are used in the wind industry today:

Doubly-Fed Induction Generator (DFIG). Used by manufacturers such as Nordex, Vestas, and Siemens Gamesa (onshore). The rotor is connected to the grid through a partial-scale power converter (handling only 25–35% of rated power), while the stator is connected directly. This results in a smaller, cheaper converter and high efficiency at rated power. However, the direct stator–grid coupling makes the machine more sensitive to grid voltage fluctuations, and the gearbox and slip rings require more frequent maintenance.

Gearless Direct Drive (Permanent Magnet Synchronous Generator – PMSG). Used by Enercon and Siemens Gamesa (offshore). The rotor shaft drives the generator directly, eliminating the gearbox. A full-scale power converter decouples the generator entirely from the grid, enabling superior grid performance (fault ride-through, precise active and reactive power control). The absence of a gearbox reduces maintenance requirements and potentially extends the turbine’s operational lifetime. The trade-off is a larger, heavier, and more expensive full-power converter chain, and more complex fault diagnostics. Enercon’s design has evolved over generations: older models (e.g. E82) placed the rectifier in the nacelle and the inverter at the tower base via a DC link, while current models (E115) integrate the full conversion chain in the nacelle, and newer large models (E175) consolidate all power electronics in a compact nacelle structure.

› 3.1.5 Construction

Logistics and Transport

Wind turbine transportation is one of the most challenging aspects of onshore wind projects in densely populated regions such as Belgium. Components – blades (approximately 65 m), tower sections, and nacelles (approximately 13 m long, 4 m wide) – are manufactured in Europe and transported by sea to the port of Antwerp. A local specialised transport company then takes over responsibility for the convoy from the port to the site, including liability for damage or blockages.

The transport company conducts detailed route surveys, produces turn simulations using dedicated software, and performs dummy runs (typically at night) to validate the route before actual component delivery. Road adaptations may include removing road signs, applying steel plates on roundabout islands, and using high-density polyethylene (PEHD) ground protection plates to extend turning radii on agricultural land. From the road transition point (RTP) onwards, Ventis assumes full responsibility for road design, land agreements, and any delays or damage.

Civil Works (Balance of Plant)

Access roads and crane platforms must meet strict bearing capacity requirements: up to 12 t per axle and a total crane load of up to 137 t. Road layers are designed using finite element modelling (FEM) based on geotechnical reports and OEM specifications, and validated on-site by plate load tests (EV1/EV2 compaction ratios). Water management (drainage, avoiding mud and icing) is also a critical design criterion.

Foundations. Two main categories of foundation design are used: slab (or gravity) foundations and pile foundations. The choice depends on geological conditions (pile foundations are unsuitable for karstic limestone), available land area, groundwater level (foundations are elevated to reduce buoyancy and increase hub height), and cost (depth to competent rock is a key driver). A geotechnical campaign – comprising penetration tests, core drilling, electrical resistivity tomography, and piezometer installation – is mandatory to characterise the subsoil. The Gouy-Lez-Piétons foundations have a diameter of 19.2 m, compared to 17.6 m for a comparable 2 MW turbine from 2007, reflecting the increased loads of modern larger machines.

Gravity foundation variants include direct soil replacement and soil improvement techniques (ballasted columns, rigid inclusions, or mixed inclusions). Pile foundations may use driven piles (steel or concrete, impact-driven) or bored piles (various auger techniques). Concreting of the slab is performed in a single continuous pour, and the foundation is then backfilled and compacted.

Electrical Infrastructure

MV cables (24 kV, aluminium, cross-sections of 150 or 240 mm²) are buried at a depth of approximately 1.3 m, laid in a bed of sand with PE protection slabs and warning ribbons, alongside HDPE conduits for fibre-optic communication cables.

The on-site electrical cabin contains two distinct sections. The high-voltage (HV) side houses ten SF₆ switchgear cells: three for ENGIE (departure, metering, generator protection), four for ORES (two arrival feeders, reference voltage, ORES protection), and three for Ventis/GOUYNRG (generator protection, metering, departure), plus an auxiliary transformer. The low-voltage (LV) side contains the SCADA controllers for each owner (Nordex

CWE units), an aggregator box for curtailment management based on electricity prices, a NORTEC unit for automated shadow flicker and bat curtailment, a Phoenix UPS, internet routers with fixed IPs and firewalls, and a 4G backup system.

Turbine Assembly and Commissioning

Assembly involves the positioning of a main crawler crane (LTR1100) and an assist crane (LTM1650-8.1) according to OEM lifting plans. Tower sections, nacelle, hub, and blades are lifted and assembled sequentially. Commissioning includes electrical completion, fibre-optic cabling, functional test runs, grid energisation, and formal inspections by the DSO and a certified independent body (e.g. TÜV, Vinçotte, Bureau Veritas). A punch list process formalises the handover between the OEM and the owner.

‣ **3.1.6 Operation and Maintenance**

Operation of the wind farm encompasses preventive maintenance, corrective maintenance, remote monitoring, mandatory regulatory inspections, and environmental monitoring.

Preventive maintenance includes periodic bolt re-torquing (tower flanges, nacelle, blade roots), regreasing of gearboxes, pitch bearings, and yaw rings, visual inspections of blades, nacelle components and tower internals, functional testing of safety systems, and planned replacement of wear parts. A condition monitoring system (CMS) using vibration sensors on the main bearing and gearbox allows early detection of anomalies such as under-greased bearings, enabling intervention before catastrophic failure.

Corrective maintenance covers unplanned repairs such as blade damage from lightning strikes (repaired by rope-access technicians) and major component replacements such as generator swaps, which require a dedicated service crane.

Remote monitoring is performed via the Farsight SCADA platform, which provides real-time dashboards, power curve analysis, wind analysis, alarm management, and reporting for each turbine across the portfolio.

Environmental monitoring obligations include post-construction bat and bird mortality surveys (using trained dogs and regular transect walks), shadow flicker management (automated curtailment via NORTEC when a receptor exceeds the permitted annual shadow hours), noise measurements at neighbouring dwellings, and ice detection on blades.

‣ **3.1.7 Grid Constraints and Power Offtake Optimisation**

Curtailments

The 25 MW injection capacity agreed with ORES limits the farm's ability to export its full 28.8 MW installed capacity at all times. Additionally, curtailments are triggered when day-ahead or intraday electricity market prices turn negative, since injecting power at negative prices generates a direct financial loss. Environmental curtailments (bats, shadow, noise) add further stops, typically representing 4% of gross annual production at Gouy.

Grid Congestion Context

The Belgian distribution grid is under significant stress: renewable energy connection requests to ORES increased by more than a factor of four between recent years, and the total demand connected or pending connection is growing rapidly (from approximately 350 MW measured load to projections above 500 MW by 2025), driven by battery storage, EV fast charging, and industrial electrification. The grid is aging and was designed for a unidirectional power flow from large central plants to consumers – the opposite of distributed renewable generation.

‣ **3.1.8 Future Developments**

Ventis identifies four main optimisation vectors to improve the economic viability of its portfolio under increasing grid congestion and market volatility:

Behind-the-meter direct connection (private PPA). A direct electrical line between a wind farm and a nearby industrial customer bypasses the distribution grid entirely, eliminating network tariffs. Both parties benefit: the producer receives a higher price than the spot market, and the industrial customer pays less than the regulated tariff. This arrangement also tends to facilitate permitting, as industrials committing to green energy consumption are viewed favourably. Battery storage can further improve the self-consumption rate.

Repowering. End-of-life turbines are replaced by fewer but larger and more powerful machines, making use of the existing grid connection, partially secured land, and an already-integrated landscape. Modern turbines deliver significantly more energy per unit cost than those installed 20 years ago. Permitting is generally more straightforward, as the visual impact on the landscape is already accepted by the local community and the turbine count decreases.

Battery Energy Storage Systems (BESS). BESS co-located with wind farms can absorb energy that would otherwise be curtailed (due to grid limits or negative prices) and re-inject it later. They also enable participation in ancillary service markets (frequency containment reserve FCR, automatic frequency restoration reserve aFRR, imbalance settlement) and support peak-shaving strategies. One concrete project under development involves a 2 MW / 4 MWh BESS at a 10.8 MW wind farm (PEDG) with an 8.8 MW injection limit.

EV fast-charging stations. Wind turbines can supply EV chargers through a direct private line, providing traceable renewable electricity for mobility without loading the public grid. This creates a win-win situation for the producer (better selling price) and the consumer (cheaper green electricity), reduces the need for municipalities to fund public charging infrastructure, and increases local acceptance of associated storage or wind projects. A concept project (MWPk) envisions approximately 10 fast chargers near a motorway junction with a potential of 50 000 vehicles per day, powered by a 9.4 MW wind farm with no impact on the local distribution grid.

Wind/solar hybridisation. Solar PV generation is strongly complementary to wind in Belgium: wind output peaks in winter and at night, while solar generation peaks in summer and midday. Combining both sources at a shared grid connection point improves the utilisation of the grid capacity and the associated BESS, reduces the net cost of grid connection for the solar project, and creates opportunities for agrivoltaic development beneficial to local farmers. A project under study (EAE) combines a 25 MW wind farm with a 700 kWp solar installation and a 3.5 MW / 7 MWh BESS.

» Chapter 4 — Slow Heat

4.1 SlowHEAT – Heat the Body, Not the Building

Synthesis of the presentation by Prof. Geoffrey van Moeseke, UCLouvain, February 18, 2026

» 4.1.1 Core Intuition and Motivation

The buildings sector is one of the largest contributors to greenhouse gas emissions in Europe, and space heating accounts for the bulk of residential energy consumption. The dominant policy response has been to invest massively in building renovation — improving insulation, replacing windows, and upgrading heating systems. The EU renovation wave is estimated to require roughly €243 billion per year to achieve its climate objectives. Yet a simple thought experiment reveals the staggering scale of this commitment: equipping every EU citizen with high-quality insulating outerwear would cost approximately €12.5 billion per year — nearly twenty times less.

This provocative comparison is not a call to replace renovation with ski suits, but rather an invitation to question a deeply embedded assumption: that thermal comfort necessarily means heating the entire volume of a building to a uniform temperature. The SlowHEAT project, led by Prof. Geoffrey van Moeseke at UCLouvain and conducted in collaboration with UCLouvain’s iPSY institute, the *habitat et participation* association, LAB Architecture, *Architecture et Climat*, and ULB-IGEAT, with support from Innoviris Brussels, explores a radically different paradigm. Rather than heating the building, the core idea is to direct heat toward the occupant’s body through *Personal Comfort Systems* (PCS) — a concept with deep historical roots and a growing body of scientific support.

» 4.1.2 An Idea Rooted in History

The notion of localised, body-centred heating is far from new. For millennia, heat was confined to specific spots within the dwelling. Ancient Greek braziers, Gaulish central hearths, Viking longhouse fire pits — all provided radiant warmth to those gathered around them, while leaving the rest of the space cold. Furniture was similarly designed to concentrate warmth: canopied and curtained beds created insulated sleeping micro-environments, hooded chairs shielded occupants from draughts, folding screens blocked cold air, and small portable foot warmers carried embers beneath one’s skirts. Japanese kotatsu tables, combining a low table with a heating element and a heavy draping blanket, are a particularly elegant example of the same principle: heat the body, not the room.

The shift toward uniform centralised heating began in earnest in the late nineteenth and early twentieth centuries, with the industrialisation of cast-iron radiators and gas or coal boilers. This transition was not purely technological — it was also cultural. Advertising campaigns of the 1930s (including those of the Compagnie Nationale des Radiateurs in France and the National Radiator Company in England) actively constructed a new narrative: central heating was modern, hygienic, uniform, effortless, and liberating. The old way of warming — gathering around a stove, layering clothing indoors, retreating to a heated corner of the house — was reframed as backward, uncomfortable, and redolent of poverty.

The consequences have been significant. Historical data show that average indoor winter temperatures in UK homes rose from approximately 15–16 °C in the 1960s to 19–20 °C by the late 1990s. Research by Mavrogianni et al. (2013) documents this steady upward drift across living rooms, bedrooms, and hallways alike. At the EU scale, the European Environment Agency has estimated that switching from single-room heating to central heating increases energy demand for space heating by roughly 25 % on average — an increase that has partly offset the efficiency gains achieved through better insulation and more efficient boilers over the same period.

› 4.1.3 An Idea Aiming for Progress

Far from being a nostalgic regression, the SlowHEAT approach is positioned as a forward-looking response to multiple contemporary challenges.

Improved health. Recent work published in *Energy & Buildings* (Zhu et al., 2025) underscores that prolonged exposure to thermoneutral environments may compromise both comfort and health. The human body is equipped with a sophisticated array of thermoregulatory mechanisms — vasoconstriction and vasodilation, piloerection, brown adipose tissue activation, and shivering thermogenesis — that are simply not engaged when indoor temperatures are maintained at a constant 20–22 °C. Mild, repeated cold exposure has been associated with metabolic benefits, including activation of brown fat, as demonstrated by van der Lans et al. (2013). The hormesis principle — whereby moderate physiological stress produces adaptive benefits — may apply to thermal environments just as it does to exercise.

Improved individual wellbeing. Unlike conventional HVAC systems that impose a single thermal setpoint on all occupants of a shared space, PCS allow individuals to modulate their own thermal environment without affecting those around them. Research on office environments (Wegertseder-Martinez et al.) confirms that this individualised control improves both comfort satisfaction and perceived wellbeing.

A pathway toward electrification. The phase-out of fossil fuel heating is a central objective of European climate policy. In Belgium, oil boilers are already banned in new constructions in Wallonia (from January 2026), and Brussels has extended the ban to gas boilers as well. Heat pumps are the primary proposed alternative, but their high upfront cost and the need for low-temperature distribution systems make them challenging to retrofit. Low-power electric PCS — consuming tens to a few hundred watts rather than several kilowatts — represent a more immediately accessible bridge technology, compatible with existing electrical infrastructure.

Technological and social pluralism. The approach does not prescribe a single solution. It accommodates both high-tech options (battery-powered heated garments, smart infrared panels, app-controlled electric blankets) and low-tech ones (well-chosen woollen underlayers, homemade heated capes fabricated in community repair workshops, draught exclusion). This breadth means it can appeal to both techno-optimists and advocates of degrowth and sufficiency.

› 4.1.4 The SlowHEAT Living-Lab Experiment (2020–2023)

General Framework

The empirical core of SlowHEAT is a three-year living-lab study conducted with thirty volunteer households in Belgium, running from winter 2020–2021 through winter 2022–2023. The study period coincided, unplanned, with the 2021–2022 energy price crisis triggered by the Russian invasion of Ukraine — an external shock that adds both complexity and ecological validity to the results.

Participants were voluntary, self-selected, and environmentally motivated — a sample the authors themselves acknowledge as likely to produce over-optimistic results compared to the general population. The group was predominantly highly educated (64 % held a master’s degree, 14 % a PhD), with a balanced gender distribution and a spread of age groups from the 20s to the 60s, with the largest cohort in their 40s (31 %). Housing types and energy performance levels were deliberately varied.

The methodological framework drew on social practice theory, following Schatzki (1997) and Gram-Hanssen (2010), analysing heating behaviour along four interacting dimensions: material infrastructures and objects, practical knowledge and skills, representations and meanings, and collective norms and rules. Participants were not asked to endure discomfort: the goal was to find a personal equilibrium, supported by a toolkit of PCS made available to all households. These ranged from heated bed covers (100 W) and heating chair pads (30 W) to small infrared panels (300 W), radiant heaters (1 200 W), large infrared panels (400 W), and heated desk pads (80 W).

Quantitative Results

The central quantitative findings are striking. Over the course of the project, participants progressively reduced their indoor temperatures well below conventional norms. Table 4.1 summarises the evolution of living-room temperatures.

In terms of energy consumption, participants achieved average reductions of approximately 50 % in gas consumption on a heating-degree-day adjusted basis, and approximately 14 % in electricity consumption. The authors note two important caveats: gas savings may be overestimated because most dwellings are apartments

Table 4.1: Indoor living-room temperature: thermostat setpoints before the project and measured temperatures during the project. Medians, means, and standard deviations are calculated for the whole participant group. Both occupied and unoccupied periods are included.

Period	Metric	Median (°C)	Mean (°C)	SD (°C)
Before project	Thermostat setting	19.0	19.0	1.6
First winter	Coldest week	16.7	17.6	1.9
	Coldest day	15.8	16.9	2.1
Third winter	Coldest two months	15.0	15.1	1.9
	Coldest month	14.5	14.7	2.0
	Coldest week	13.7	13.5	2.6
	Coldest day	12.5	12.6	2.5

or semi-detached houses that benefit from heat transfers from neighbouring units; and the electricity savings were unexpected, likely reflecting the broader behavioural response to the 2022 energy price shock rather than the PCS alone.

Inter-household variation was considerable. Mean indoor temperatures across participating dwellings ranged from approximately 11 °C to 16 °C, reflecting wide differences in housing performance, household composition, and personal thermal preferences. One particularly illustrative case is that of a household that had already undergone a building renovation: over the same period, SlowHEAT practices generated savings of 700 m³ of gas per year, compared to 380 m³ per year attributable to the renovation itself.

► 4.1.5 Qualitative Analysis: A Social-Practice Framework

Changing Representations

Perhaps the deepest shift required by the SlowHEAT approach is representational. Our current expectation that a home should be uniformly heated to 19–22 °C is not a biological necessity but a historically and socially constructed norm — one that, as historian Olivier Jandot argues, is inseparable from the technological evolution of the nineteenth and twentieth centuries. This norm is recent, geographically variable, and therefore changeable.

One of the most striking indicators of this normative rigidity is the public reaction to media coverage of the SlowHEAT project. Social media comments cited in the presentation include accusations of endangering children, comparisons to medieval living conditions, and assertions that living below 20 °C is simply unacceptable in 2022. These reactions illustrate the depth of the cultural embedding of centralised warmth as a marker of dignity and modernity — an embedding that was itself deliberately constructed by the heating industry a century ago.

Alongside this cultural shift, participants were invited to develop a richer vocabulary for bodily thermal sensations and to develop awareness of their body’s own thermoregulatory capacities: vasoconstriction, brown adipose tissue activation, and the adaptive reduction in perceived cold discomfort with regular mild exposure.

Acquiring Knowledge and Skills

A significant part of the learning process concerned the relationship between indoor temperature, relative humidity, and air quality. A key practical insight is that, at 15 °C, maintaining indoor relative humidity below approximately 55 % (corresponding to a humidity ratio of roughly 0.006 g_{water}/kg_{air}) is sufficient to prevent mould growth on most building materials. Critically, in a Belgian winter climate, outdoor air humidity rarely exceeds this threshold, meaning that regular ventilation with outdoor air is almost always safe from a mould-prevention perspective — a counterintuitive finding for many participants who feared that cooler interiors would be moister and more prone to mould.

The SlowHEAT team also developed a dedicated tool, the *SlowZone* widget, which combines real-time outdoor temperature and humidity data with psychrometric calculations to indicate the safe ventilation window for any given location.

Participants also developed practical skills in selecting and deploying PCS: understanding the difference between conductive heating (body contact), radiant heating (infrared proximity), and convective heating (air warming), and learning which solution best matched each activity, body zone, and time of day.

Infrastructure and Technology: A Hierarchy of Solutions

The project proposes a structured hierarchy of heating interventions, ordered from least to most energy-intensive, as summarised in Table 4.2.

Table 4.2: Proposed hierarchy of Personal Comfort System interventions, from zero-energy measures to whole-building heating.

Class	Power	Solution family	Examples
A – Zero-energy measures	0 W	Clothing, zoning, activity	Thermal underwear, closing doors, physical activity, curtains
B – Body-contact conduction	± 50 W/person	Wearable or contact accessories	Heated chair pad, electric blanket, heated cape, hot-water bottle
C – Radiant proximity	± 300 W/person	Radiant or heated furniture near occupant	Infrared panel, radiant column, heated table
D – Local room heating	$\pm 1\,500$ W/room	Thermostatic room heater	Room maintained at 15–17 °C when occupied, with Classes A–C still active
E – Whole-building heating	$\pm 5\,000$ W/dwelling	Central heating system	Kept at frost-protection level (8 °C) or passage temperature (12–15 °C)

The philosophy is not to eliminate central heating entirely, but to relegate it to a supporting, background role while Classes A through C meet most comfort needs. Low-tech Class A and B solutions are emphasised: the project ran community sewing workshops to produce washable heated capes equipped with recycled heating textile, and explored prototype heated chair inserts and desk setups using low-voltage heating elements.

Navigating Social Norms

The social dimension proved one of the most challenging aspects of the experiment. Participants navigated real tensions between their new practices and the expectations of visitors, family members, and social convention. Receiving guests in a 15 °C home, asking in-laws to put on a jumper, or negotiating thermostat settings within a household all require a form of social courage and communication that purely technological solutions do not demand.

Some participants experienced these tensions positively, as a sign of being ahead of their time. Others found the social cost significant. The project also surfaced a regulatory tension: current energy performance certificate frameworks in Wallonia (and more broadly across Europe) rate buildings on the basis of normalised energy consumption at standard occupancy and temperature assumptions. A household that genuinely heats its home to 14 °C and achieves excellent comfort through PCS may appear, on paper, to have a poorly performing building — or may even find that certain renovation subsidies are inaccessible because its actual consumption already falls below the required thresholds. The authors argue that regulation should distinguish more clearly between the *means* (energy in kWh/m²) and the *end* (a healthy, enabling domestic environment), and that “off-framework” renovation pathways tailored to sober lifestyles should be legally recognised.

► 4.1.6 Discussion: Efficiency, Sufficiency, and Beyond

SlowHEAT raises fundamental questions about the relationship between efficiency and sufficiency in energy policy. The dominant paradigm — enshrined in frameworks such as the Negawatt approach — places sufficiency first, followed by efficiency, then renewable energy. In practice, however, sufficiency is rarely operationalised with the same rigour as efficiency: it tends to mean marginal reductions (lower the thermostat by 1 °C, reduce shower time by two minutes) rather than a qualitative transformation of how services are delivered.

SlowHEAT illustrates what a more radical form of sufficiency can look like in practice: not “less of the same” (a slightly cooler uniformly heated home) but a fundamentally different service configuration (a cool home with locally warm people). This distinction is important not only for its energy implications but for its social ones. As the concept of *consumption corridors* (Akenji et al., 2021) suggests, sufficiency should be bounded both by a social floor — below which underconsumption causes deprivation and health risk — and an environmental ceiling — above which overconsumption is ecologically unsustainable. Both boundaries should be democratically

established and informed by science. The SlowHEAT experiment suggests that the social floor for thermal comfort may be considerably lower than current norms imply, provided adequate PCS are available and occupants have the knowledge, agency, and social support to use them.

The project's main acknowledged limitations are twofold. First, the sample's high educational level and prior environmental motivation make it difficult to generalise: a participant group with mixed motivations and lower climate awareness might achieve smaller reductions and report lower satisfaction. Second, the energy savings attributed to SlowHEAT may be partially overestimated in multi-unit dwellings, where heat transfers from neighbouring heated apartments contribute to maintaining indoor temperatures. More controlled studies, with matched comparison groups and sub-metered heating data, are needed to isolate the specific contribution of PCS from confounding behavioural and contextual factors.

Notwithstanding these limitations, the SlowHEAT project makes a compelling empirical case that a significant fraction of European households could, with appropriate support, substantially reduce their heating energy consumption while maintaining subjective comfort — and that the barriers to doing so are as much cultural, regulatory, and social as they are technical.

» Chapter 5 — Batteries

5.1 Introduction: Energy Storage and Electrochemistry

The growing demand for portable energy, electric vehicles, and renewable energy integration has made efficient energy storage a central challenge of our time. Raw energy — whether from fossil fuels, nuclear reactions, solar irradiation, or wind — must often be transformed and stored before use. Among storage technologies, electrochemical batteries offer a unique combination of portability, scalability, and high energy conversion efficiency.

The classical thermomechanical route to electricity — combustion, thermal engine, alternator — is fundamentally limited by the Carnot principle, which caps efficiency at $\eta_{\max} = 1 - T_c/T_h$. The electrochemical route, by contrast, achieves a direct conversion of chemical energy into electrical work, with a theoretical yield given by:

$$\eta_{\text{th}} = \frac{\Delta G}{\Delta H} \quad (5.1)$$

For a hydrogen/oxygen fuel cell, this yields $\eta_{\max} \approx 95\%$, far exceeding any heat engine operating between practical temperature limits.

The fundamental unit of an electrochemical energy storage system is the **electrochemical cell**, composed of two electrodes separated by an electrolyte:

$$\text{electrode} / \text{electrolyte} / \text{electrode}$$

At constant temperature and pressure, the Gibbs free energy of the cell reaction is directly related to the cell voltage:

$$\Delta G(T, P) = -nF\Delta\varepsilon \quad (5.2)$$

where n is the number of electrons exchanged, $F = 96,485 \text{ C/mol}$ is Faraday's constant, and $\Delta\varepsilon$ is the electromotive force. The spontaneity condition requires $\Delta\varepsilon > 0$. The maximum achievable cell voltage is:

$$\Delta V_{\max} = \Delta\varepsilon = -\frac{\Delta G}{nF} \quad (5.3)$$

For example, the lead-acid cell gives $\Delta V_{\max} = 2.1 \text{ V}$ per cell, so a standard 12 V car battery consists of six cells in series.

Rechargeable (secondary) batteries must additionally satisfy three conditions: the cell reaction must be reversible, the charge/discharge cycling yield must be acceptable, and material degradation must remain within tolerable bounds.

The history of rechargeable batteries spans nearly two centuries, from Planté's lead-acid cell (1859) through nickel-cadmium and nickel-metal hydride technologies, culminating in the lithium-ion battery developed between 1970 and 1990 by Whittingham, Goodenough, and Yoshino — work recognised by the 2019 Nobel Prize in Chemistry. Over this period, the specific energy of rechargeable batteries has risen from roughly 30 Wh/kg (lead-acid) to over 180 Wh/kg (Li-ion), with volumetric energy density approaching the theoretical limit of around 400 Wh/L.

5.2 How a Li-ion Battery Works

› 5.2.1 General Operating Principle

A lithium-ion battery operates through the **reversible intercalation** of lithium ions (Li^+) between two host electrode materials. Crucially, lithium is never reduced to its metallic form during normal operation; it remains ionic throughout the charge/discharge cycle.

- **Negative electrode (anode):** graphite, which hosts Li^+ ions in its layered structure to form LiC_6 .
- **Positive electrode (cathode):** a lithiated transition metal oxide, typically of the form LiMO_x (e.g., LiCoO_2 , LiFePO_4 , LiMn_2O_4 , or NMC variants).

The overall cell reaction can be written as:



During **discharge**, Li^+ ions de-intercalate from the graphite anode, travel through the electrolyte, and re-intercalate into the cathode oxide, while electrons flow through the external circuit producing electrical work. During **charge**, the process is reversed.

› 5.2.2 Criteria for Electrode Materials

An effective intercalation electrode material must satisfy several criteria simultaneously:

1. **High Li^+ insertion capacity:** the material must accommodate a large number of lithium ions per formula unit, yielding high specific capacity (mAh/g).
2. **Reversibility:** insertion and de-insertion must be highly reversible with minimal hysteresis, ensuring long cycle life.
3. **Voltage plateau:** the electrode potential should be stable (flat plateau) at an adequate value with respect to Li^+/Li .
4. **Good electronic and ionic conductivity:** both electron and ion transport through the electrode must be fast enough to support practical current densities.
5. **Low volume change:** the crystal structure should expand and contract minimally upon Li^+ insertion/extraction to avoid mechanical degradation.
6. **Durability and low cost:** the material must be stable over hundreds to thousands of cycles and manufacturable from abundant, inexpensive precursors.

The stored energy of a cell is given by:

$$E = \Delta U \times i \times t \quad (5.5)$$

where ΔU is the electromotive force (difference between cathode and anode potentials) and $i \times t$ is the capacity delivered.

› 5.2.3 Representative Electrode Materials

Table 5.1 summarises the key properties of common positive and negative electrode materials.

Li-ion cells typically deliver 3.5–3.8 V, compared to 2.0 V for lead-acid and 1.2 V for Ni-MH cells. Compared to competing rechargeable technologies, Li-ion offers a superior energy density (180–220 Wh/kg vs. 30–80 Wh/kg for older chemistries), faster charging (60–120 min), lower self-discharge (5–10%/month), and a charge/discharge round-trip efficiency exceeding 85%.

5.3 Battery Construction

› 5.3.1 Cell Architecture

The electrochemical cell consists of thin-film layers stacked or wound together:

current collector | anode | electrolyte | separator | electrolyte | cathode | current collector

Table 5.1: Properties of selected Li-ion electrode materials.

Material	Avg. voltage	Specific capacity	Mass energy
<i>Positive electrodes</i>			
LiCoO ₂	3.7 V	140 mAh/g	0.518 kWh/kg
LiMn ₂ O ₄	4.0 V	100 mAh/g	0.400 kWh/kg
LiFePO ₄	3.3 V	150 mAh/g	0.495 kWh/kg
<i>Negative electrodes</i>			
Graphite (LiC ₆)	0.1–0.2 V	372 mAh/g	0.037–0.074 kWh/kg
Titanate (Li ₄ Ti ₅ O ₁₂)	1–2 V	160 mAh/g	0.16–0.32 kWh/kg
Si (Li _{4.4} Si)	0.5–1 V	4212 mAh/g	2.1–4.2 kWh/kg

The anode current collector is copper foil; the cathode current collector is aluminium foil. The separator is a porous polymer membrane that prevents electronic contact between the two electrodes while allowing ionic transport.

› 5.3.2 Composite Electrodes

Practical electrodes are not monolithic but **composite structures** deposited as thin films (typically 100 µm thick) on the metallic current collector. Each electrode layer contains:

- **Active material** (powder): the intercalation compound itself.
- **Binder**: typically polyvinylidene fluoride (PVDF), dissolved in N-methylpyrrolidone (NMP) to form a slurry. PVDF is chemically stable but is a fluorinated polymer and therefore environmentally persistent and difficult to recycle. Alternative water-based binders (cellulose-derived gums, carboxymethyl cellulose) are actively being developed.
- **Conductive additive**: usually carbon black, to ensure electronic percolation through the electrode since many active materials have low intrinsic electronic conductivity.

The assembled electrode is soaked with liquid electrolyte, which fills the porous space between particles and ensures ionic access to all active surfaces. Nanostructuring the active material (reducing particle size, introducing porosity) is a key strategy to maximise accessible surface area and reduce solid-state diffusion distances for Li⁺.

› 5.3.3 Electrolyte

The electrolyte must be a pure ionic conductor (electron insulator) that is stable across the full voltage window of the cell (~4 V for standard Li-ion). Two main types are used:

- **Liquid electrolyte** ("classical" Li-ion): a lithium salt (typically LiPF₆) dissolved in a mixture of organic carbonate solvents (e.g., ethylene carbonate, dimethyl carbonate). These electrolytes are highly flammable, requiring the addition of flame-retardant additives and imposing strict purity requirements during manufacture.
- **Gel polymer electrolyte** (Li-ion "polymer"): a polymer matrix swollen with liquid electrolyte, offering improved safety and mechanical flexibility.

A critical limitation of liquid electrolytes is that their ionic conductivity decreases significantly at low temperatures, degrading battery performance in cold climates.

› 5.3.4 Cell Formats and Assembly

Because a single cell provides only 3.5–3.8 V, practical battery packs assemble many cells in series (to increase voltage) and in parallel (to increase capacity). The cell format can be cylindrical (wound jelly-roll), prismatic (stacked layers), or pouch (flat flexible). Battery manufacturing is a complex multi-step process involving slurry mixing, coating, drying, calendaring, slitting, winding or stacking, electrolyte filling, and final sealing, all under tightly controlled atmospheric conditions to exclude moisture and metallic contaminants. Even microscopic metal particles can cause internal short circuits, thermal runaway, and fire.

At the pack level, a **Battery Management System (BMS)** is essential to monitor cell voltages, temperatures, and state of charge, and to balance cells, prevent overcharge/overdischarge, and manage thermal conditions.

5.4 Challenges

› 5.4.1 Materials Challenges

The development of Li-ion batteries relies on several materials that are classified as **critical raw materials** under EU legislation, including lithium, cobalt, nickel, manganese, and natural graphite. These elements are geographically concentrated, subject to supply chain risks, and often associated with environmental and ethical concerns in mining. This drives intensive research into alternative chemistries.

Silicon is a particularly attractive anode candidate, with a theoretical capacity of ~ 4000 mAh/g compared to only 372 mAh/g for graphite. However, Si undergoes a volume expansion of $\sim 320\%$ upon full lithiation (to $\text{Li}_{22}\text{Si}_5$), causing mechanical cracking and loss of electrical contact upon cycling. Strategies to mitigate this include nanostructured silicon (nanoparticles, nanowires, honeycombs, micropillars), porous composite architectures embedding Si nanoparticles in a compliant carbon matrix, and silicon-graphite blended anodes already entering commercial production.

On the cathode side, cobalt-free or cobalt-reduced NMC ($\text{LiNi}_x\text{Mn}_y\text{Co}_z\text{O}_2$) formulations with high nickel content (NMC811, NMC90) are being pursued to reduce cost and supply risk while improving energy density. LiFePO_4 (LFP) has re-emerged strongly as a safe, low-cost, cobalt-free cathode with excellent cycle life, particularly for stationary storage and lower-range EVs.

› 5.4.2 Manufacturing and Component Challenges

Component-level challenges include improving electrode adhesion, electronic conductivity, and electrolyte compatibility. The replacement of PVDF/NMP binder systems with water-based alternatives (biopolymer gums such as xanthan or carboxymethylcellulose) is an active area, motivated both by environmental concerns and by the need to reduce production costs and eliminate solvent recovery steps.

At the battery assembly level, the absence of any metallic contamination is a paramount constraint, since even sub-millimetre metal particles can provoke internal short circuits leading to thermal runaway. Clean-room manufacturing environments, continuous monitoring, and automated optical inspection are indispensable.

› 5.4.3 Cost Trajectory

The price of Li-ion battery packs has fallen dramatically: from approximately \$780/kWh in 2013 to around \$139/kWh in 2023. The widely cited cost target of \$100/kWh is considered the threshold at which electric vehicles reach purchase price parity with internal combustion vehicles without subsidies. Continued cost reduction requires simultaneously improving energy density, reducing reliance on expensive materials, and scaling manufacturing.

› 5.4.4 Safety, Recycling, and End-of-Life

Safety remains a concern due to the flammability of liquid electrolytes and the risk of thermal runaway. Solid-state electrolytes — ceramics, sulfides, or polymer-ceramic composites — promise intrinsically safer batteries by eliminating flammable liquids and enabling lithium metal anodes, but achieving sufficient ionic conductivity and stable electrode/electrolyte interfaces at room temperature remains an unsolved challenge.

Recycling is increasingly critical both for resource recovery and for reducing the environmental footprint of battery production. Current recycling routes (pyrometallurgy, hydrometallurgy, direct recycling) recover metals to varying degrees but remain costly and energy-intensive. The EU Battery Regulation introduces mandatory recycled content targets and collection obligations.

5.5 Future Perspectives

› 5.5.1 Beyond Li-ion: Sodium-Ion Batteries

Sodium-ion (Na-ion) batteries follow the same intercalation principle as Li-ion but replace Li^+ with Na^+ . Sodium is vastly more abundant and geographically distributed than lithium, and Na-ion cells can be assembled without copper current collectors (since sodium does not alloy with aluminium at low potentials, aluminium can be used for both electrodes). Water-based electrode manufacturing is also feasible. Current Na-ion cells reach around 100 Wh/kg with cycle lives exceeding 2000 cycles — lower than state-of-the-art Li-ion but improving rapidly. Cathode materials include sodium vanadium fluorophosphates (NaVPO_4F and variants); hard carbon is the

preferred anode. Industry is moving fast: BYD announced in February 2026 a third-generation Na-ion pack targeting 10,000 cycles, alongside a solid-state battery programme for 2027 EVs.

› 5.5.2 Solid-State Batteries

Solid-state batteries replace the liquid electrolyte with a solid ionic conductor (oxide, sulfide, or polymer-based). They promise higher energy density (by enabling lithium metal anodes), intrinsic safety (no flammable liquid), and potentially wider operating temperature ranges. Key unresolved challenges include: achieving room-temperature ionic conductivity comparable to liquid electrolytes ($\sim 10^{-2}$ S/cm), ensuring mechanically stable and low-resistance electrode/electrolyte interfaces over thousands of cycles, and scaling up manufacturing.

› 5.5.3 A Broader Perspective

Battery technology is ultimately only one part of the energy transition. Research cited in the presentation (Barrett et al., *Nature Energy*, 2022) estimates that current per-capita energy consumption in developed countries averages ~ 6 kW/person, while sustainability analyses suggest a target of ~ 2 kW/person is necessary. Improving energy storage is therefore inseparable from the broader imperative to reduce absolute energy demand.

» Chapter 6 — Biomass

6.1 Introduction

Biomass — organic matter derived from living or recently living organisms — represents the oldest energy source used by humanity and remains today a central, if contested, pillar of the global energy system. This document synthesises the key technical, statistical, environmental, and philosophical dimensions of biomass as presented in a lecture on biomass and the energy transition, with particular attention to its sustainability. The core argument running through the material is that biomass itself is neither inherently sustainable nor unsustainable: what determines its environmental and social value is the system of management and exploitation within which it is embedded.

6.2 What Is Biomass?

Biomass designates any organic material of biological origin used for energy purposes. It encompasses a wide and heterogeneous range of feedstocks:

- **Forestry biomass:** stemwood, logging residues (treetops, branches), and material from thinning operations.
- **Wood industry by-products:** sawdust, bark, and chips generated during the processing of timber into lumber or panels.
- **Energy crops:** lignocellulosic crops (e.g., miscanthus, short-rotation coppice), oil crops (sunflower, palm, rapeseed), and sugar/starch crops (corn, wheat, sugar cane, sugar beet) used for bioethanol or biodiesel production.
- **Agricultural residues:** straw and other crop residues left after harvest.
- **Animal and agro-industrial residues:** manure, slaughterhouse by-products, and organic food processing waste.
- **Waste streams:** the organic fraction of municipal solid waste (40–60% of the total), demolition wood, paper and cardboard, and sewage sludge.

These feedstocks can be converted through multiple routes — combustion, transesterification, hydrolysis and fermentation, gasification, pyrolysis, and anaerobic digestion — into a variety of end-uses: heat, electricity, liquid transport fuels, and chemical feedstocks.

6.3 The Current Role of Biomass in Energy Systems

› 6.3.1 Global and European Context

Historically, traditional biomass was the dominant global energy source until the industrial revolution, after which fossil fuels rapidly took over. Today, modern bioenergy has regained importance. In the European Union, biomass is the **leading renewable energy source**, accounting for the largest share of primary renewable energy production. Nevertheless, the overall consumption mix in Europe remains heavily dominated by fossil fuels, with biomass representing a comparatively modest share of total primary energy consumption.

A similar structure is observed in Belgium, where the consumption pattern mirrors that of the EU27: fossil fuels dominate, with biomass playing a significant but secondary role among renewables.

› 6.3.2 End-Uses of Biomass in Europe

The dominant end-use of biomass in Europe is **heat production**, particularly for residential and industrial applications. Biofuels for transport represent approximately 13% of biomass use. Bioelectricity production, while growing, is a smaller share. Among the different biomass types used in the EU27, solid biomass is overwhelmingly dominant, followed at a much lower level by liquid biofuels, biogas, and renewable municipal waste.

› 6.3.3 Feedstock Composition

In Europe in 2020, **forestry products** represent by far the largest feedstock category for bioenergy (approximately 1,200 TWh), with agricultural products and other waste each contributing roughly 250 TWh. Within solid biomass, pellets represent only about 10% of the total; the remaining 90% consists of other solid biomass forms such as firewood and chips. Within biofuels, biodiesels account for 76%, bioethanol for 17%, and other biofuels for 7%.

According to a JRC (2021) analysis, the origin of woody biomass used for bioenergy in the EU (2015 data) breaks down as follows: 49% is secondary woody biomass (largely domestic solid by-products at 21%, and black liquor at 15%), while 37% is primary wood. Of this primary wood, roughly 47% is stemwood and 53% consists of other forest components such as treetops and branches. Additionally, 24% is fuelwood and 4% industrial roundwood. This means that a substantial part of wood used for energy is a co-product of forestry management operations, including thinnings, rather than wood harvested primarily for energy.

In summary, wood-energy in the EU is a mix of approximately 50% wood extracted directly from forests (largely as a co-product of timber forestry management) and 50% by-products from the wood processing industry. Sawing logs typically yields only 45–60% of usable timber volume, leaving a large stream of residues available for energy use.

6.4 Sustainability of Biomass

› 6.4.1 The Debate on Liquid Biofuels

Liquid biofuels — particularly first-generation biofuels based on food and feed crops — have been at the centre of significant controversy. The Belgian case of Biowanze illustrates the complexity: this facility processes around 800,000 tonnes of wheat annually (50% from Belgium, 30% from France, 10% from Germany, 10% from other EU countries) to produce bioethanol alongside food-grade co-products (proteins, liquid CO₂) and green energy.

Life cycle assessment (LCA) of Belgian wheat bioethanol shows that compared to gasoline (per MJ of energy delivered), bioethanol performs better on **climate change** (approximately 0.04 kg CO₂eq vs. 0.084) and on fossil fuel depletion. However, it performs worse on **particulate matter formation, acidification, terrestrial and freshwater eutrophication, land use, and mineral depletion**. The verdict therefore depends critically on which environmental impact categories are prioritised.

The Energy Return on Energy Invested (EROI) — defined as the ratio of energy output to energy input across the life cycle — provides a complementary physical lens. The EROI of first-generation bioethanol is particularly low, often close to 1. For wheat-based bioethanol, reported values range from 0.69 to 2.31 depending on system boundaries, co-product allocation methods, and regional conditions. This places corn- and wheat-based ethanol near the bottom of the so-called "net energy cliff," far below fossil fuels (EROI ~20–100 historically) and below other renewables such as wind and photovoltaic energy.

EU policy has responded to these concerns: the revised Renewable Energy Directive established decreasing caps for first-generation biofuels (from 7% in 2021 to 3.8% in 2030), specifically to limit emissions linked to indirect land-use change.

› 6.4.2 The Carbon Cycle of Forests

A critical issue for the sustainability of woody biomass is the accounting of carbon stocks. In EU27 forests (2020 data), carbon is distributed as follows: 37% in above-ground biomass, 10% in below-ground biomass, 2% in deadwood, 7% in litter, and **44% in soil**. This means that the majority of forest carbon is not in the trees themselves but in the soil, a stock that is particularly sensitive to management practices and residue extraction.

At the scale of an individual tree stand, harvesting inevitably reduces the carbon stock. However, a forest is not managed at the stand scale. At the landscape scale, cutting trees in one location while replanting and managing other areas over the long term can stabilise or even increase total carbon stocks. The sustainability of carbon management therefore depends fundamentally on how the forest is managed — rotation length, residue extraction rates, species composition, and regeneration practices — rather than on the simple fact of harvesting.

› 6.4.3 The Management and Exploitation System

The lecture's central thesis is encapsulated in the following framing: it is not biomass itself that is or is not sustainable — it is the **system of management and exploitation**. This invites four key questions:

1. What system of forestry or agricultural management is currently in place?
2. What are the risks to sustainability under current logics of productivity and profit?
3. What alternatives exist, and how can coherent changes be implemented?
4. Are we even asking the right questions?

6.5 Biomass Potential and Its Variability

Estimates of the sustainable bioenergy potential of Europe vary enormously across studies, depending on the type of potential considered (theoretical, technical, economic, ecologically sustainable, or implementable) and on key assumptions:

- Dietary choices and associated land availability for energy crops;
- Agricultural landscape structure and scale of biogas facilities;
- Degree of mechanisation and use of inputs (fertilisers, irrigation);
- Competing uses for biomass (food, feed, materials, biodiversity conservation) and sustainable extraction rates.

Reviewed European studies yield central estimates ranging roughly from 2,100 to 5,200 TWh/year depending on the scenario, compared to a 2019 actual production of approximately 1,540 TWh. This variability is not merely technical: each estimate embeds a specific vision of the agricultural, forestry, and waste management systems, with attendant social, economic, and environmental implications. Choosing one estimate over another in an energy system model is therefore a **political act**, not a neutral technical choice, as it shapes the resulting modelled system and the policy recommendations derived from it.

Energy system models such as EnergyScope Pathway — which integrates 28 resources, 113 conversion technologies, and 12 demand categories — show that meeting a fossil-and-nuclear-free European energy system by 2050 would require a very large scale-up of biomass alongside wind and solar. In such scenarios, woody biomass could represent around 16% of gross available energy, and wet biomass around 7%.

6.6 Broader Dimensions: Technology, Society, and the Energy Transition

› 6.6.1 The Limits of a Purely Technical Approach

A purely engineering perspective, however sophisticated, risks overlooking essential dimensions of the energy transition. As the statistician George Box noted, all models are wrong but some are useful; the key is transparency about assumptions and limitations. Energy system models reduce complex phenomena to equations, and the selection of variables and boundaries can introduce biases and mask important interactions — social, ecological, and cultural.

This limitation becomes critical when considering the scale of change required. European primary energy consumption has remained broadly stable over the past two decades, dominated by fossil fuels. Transitioning to a renewable system within 30 years appears practically unfeasible without a genuine reduction in energy demand, which in turn requires a change in mentality and not just a change in technology.

› 6.6.2 Technology Is Not Neutral

Drawing on thinkers including Lewis Mumford, François Jarrige, and Bernard Charbonneau, the lecture argues that technical choices are never socially or politically neutral. Engineers, historically servants of military and industrial power, have been oriented primarily toward the transition of tools. But as the philosopher Ivan Illich argued (cited via Geerts 2017), any energy transition is simultaneously a transition of tools *and* a transition of thinking.

› 6.6.3 Biomass and the Human–Nature Dualism

The debate around bioenergy is particularly intense, in part, because it surfaces deep cultural tensions about humanity's relationship with nature. Burning forests for energy makes visible — and locally tangible — the consequences of energy consumption, confronting communities with a sight many prefer to leave unseen. This visibility has, paradoxically, contributed to raising awareness of sustainable forestry practices and environmentalism.

The lecture draws on the philosopher Dominique Bourg's concept of *reflective monism*: the idea that humans and non-humans belong to the same living world, with the human specificity being the capacity for reflection. Under this framing, treating biomass purely as an instrumental resource for human productivity — without attending to the complex ecological interactions on which living systems depend — is antithetical to sustainability. Imposing purely productivist logics obscures the networks of relationships essential to ecosystem integrity.

Biomass thus becomes a *heuristic concept*: a lens through which to revisit the forgotten constraint of renewability, to redefine our ethical sensibility toward the non-human, and to open necessary democratic debates about what constitutes useful versus superfluous energy services. As philosopher Jonathan Parsons writes: every energy system is first and finally a social system, and a transition from one to another implies a sociopolitical transition as much as a technical one.

6.7 Take-Away Messages

The following conclusions synthesise the key points of the lecture:

- Biomass can be a genuine ally in the energy transition, but sustainable implementation is essential to avoid systems driven purely by profit and productivity.
- It is a key energy source — particularly for heat in the EU — but one subject to limits and constraints that must be respected.
- The sustainable potential of biomass is highly variable and depends on interdisciplinary parameters (ecological, social, economic, dietary); estimating it is inherently a political exercise.
- The energy transition is fundamentally a political project that requires interdisciplinary analysis and broad democratic deliberation, not only technical optimisation.
- A genuine transition requires not only a change of energy sources but a change in our conceptions of nature, energy, and the relationship between human societies and ecosystems.

» Chapter 7 — Fluxys

7.1 Overview of Fluxys: A Purpose-Driven Energy Infrastructure Company

» 7.1.1 Company Profile

Fluxys is a Belgian-headquartered multi-molecule infrastructure company operating across the full gas value chain. With approximately 1,400 employees, the group maintains a strong European presence complemented by associated companies in South America and the Middle East. Its corporate mission is encapsulated in the tagline “*shaping together a bright energy future*”, reflecting its positioning as a purpose-driven actor committed to accelerating the energy transition.

The ownership structure of Fluxys is characterised by a public-private partnership model designed to support long-term investment horizons. The principal shareholders of Fluxys Belgium include Publicgas (77.38%), Energy Infrastructure Partners – EIP (15.22%), the Federal Holding and Investment Company (3.44%), AG Insurance (1.97%), Ethias (1.31%), and personnel and management (0.67%). The Belgian State retains a golden share. Fluxys Belgium is also partly listed on Euronext (10% of shares). Fluxys holds 100% of both Fluxys Europe and Fluxys International.

Fluxys operates across the full gas value chain in Belgium, covering terminalling (regasification and liquefaction at the Zeebrugge LNG terminal), transmission, and underground storage (at Loenhout). Its Belgian network plays a structurally important role in the security of supply of Northwest Europe, acting as a transit hub connecting Norway, LNG imports, the United Kingdom, France, the Netherlands, Germany, and Luxembourg.

» 7.1.2 Belgian Gas Network: Key Figures (2025)

In 2025, total gas imports via pipeline and LNG into Belgium reached 480 TWh (up from 407 TWh in 2024), of which 147 TWh came from LNG terminals. Transit to surrounding countries amounted to 331 TWh. Domestic consumption stood at 151 TWh. The Loenhout underground storage facility has a storage capacity of 7.6 TWh, a send-out capacity of 7.25 GWh/h, and an injection capacity of 3.75 GWh/h; it reached 100% filling level at the start of August 2025. The Zeebrugge LNG terminal has a regasification capacity of 197 TWh/year (17 bcm/year), LNG storage of 566,000 m³, and sent 146.8 TWh into the network in 2025, up from 81.5 TWh in 2024.

7.2 The European Energy Context and Regulatory Framework

» 7.2.1 Geopolitical Disruptions and Supply Shifts

The Russian invasion of Ukraine in February 2022 triggered a dramatic restructuring of European gas supply. North Stream flows were halved in July 2022, reduced to 20% of capacity, and halted definitively in August 2022. The Nord Stream sabotage occurred in September 2022. As of January 2025, Russian gas flows through Ukraine have fully ceased. In response, Europe has massively scaled up LNG imports: LNG terminal throughput across Europe grew from 915 TWh in 2021 to 1,614 TWh in 2025. Norwegian pipeline flows have remained relatively stable at around 1,233 TWh. These shifts have made Belgium’s Zeebrugge terminal a critical entry point for European gas supply diversification.

Gas prices (TTF) spiked sharply during the 2022 crisis, reaching above 300 €/MWh, before normalising. More recent price movements have been driven by events including the end of Russian Ukrainian transit flows (January 2025), the second Trump presidency and associated geopolitical uncertainty, the Israel-Iran war (June 2025), a cold snap in Europe (January 2026), and Winter Storm Fern in Texas (January 2026).

› 7.2.2 Regulatory Framework

Gas (and hydrogen) transmission in Belgium is regulated, with Belgian legislation deriving primarily from EU law. The regulatory framework has evolved through successive EU energy packages:

- **1st and 2nd Energy Packages (1996–2004):** Liberalisation of gas markets, creation of national regulatory agencies.
- **3rd Energy Package (2009):** Third-party access, ownership unbundling, creation of ACER and ENTSOG.
- **Gas Network Codes (2013–2018):** Capacity allocation, interoperability, balancing, tariffs.
- **RePowerEU and related regulation (2022):** Phase-out of Russian gas dependency, mandatory storage filling targets, joint gas purchase platform.
- **H2 and Decarbonisation Package (July 2024):** Updated EU gas market rules and a new regulatory framework for hydrogen infrastructure, including tariffs, cost mutualization, inter-temporal cost allocation, and creation of ENNOH.

Key principles of EU gas regulation include ownership unbundling, regulated tariffs, third-party access (non-discrimination), transparency obligations (REMIT, ENTSOG Transparency Platform), network codes developed by ENTSOG and ACER, and a Ten-Year Network Development Plan (TYNDP).

› 7.2.3 Tariff Computation

Allowed revenues for regulated activities are computed as the sum of operating costs (OPEX), depreciation, and a regulated margin based on the Regulatory Asset Base (RAB). The RAB of Fluxys Belgium stands at approximately €2 billion. The regulated return is calculated via a WACC formula (currently 4.80% gross), which combines:

- ROE on the first part of equity (up to 40% of RAB), based on the CAPM model;
- ROEE on the second part of equity (above 40% of RAB), remunerated slightly above the risk-free rate;
- Cost of debt on 45% of RAB (pass-through mechanism).

Total allowed revenues amount to approximately €420 million, decomposed into €100 million return, €230 million operating costs, and €90 million depreciation. A regulatory account adjustment of –130 million is applied in the current tariff period. Unit tariffs per service are derived by dividing allowed revenues by reference quantities (Qref) for each activity (transmission, storage, LNG terminal). The tariff period is four years.

7.3 The Energy Transition: Fluxys's Strategic Role

› 7.3.1 The Energy Trilemma

As an essential infrastructure provider, Fluxys frames its energy transition strategy around balancing three imperatives: **sustainability** (reducing carbon emissions), **affordability** (keeping energy costs manageable), and **security of supply** (ensuring reliable access to energy). These three dimensions constitute the classical “energy trilemma” and guide Fluxys’s infrastructure investment decisions.

› 7.3.2 The EU Net-Zero Pathway and Belgium's Energy Mix

The EU Green Deal targets net-zero greenhouse gas emissions by 2050, starting from 3.75 Gt CO₂ in 1990. Achieving this requires a combination of solutions: energy efficiency, green electricity generation, scaling up the hydrogen economy, biomethane and synthetic methane, and carbon capture and utilisation/storage (CCUS).

Belgium's 2024 final energy consumption stood at 435 TWh, dominated by oil (48%), natural gas (25%), and electricity (17%). The Belgian electricity mix (81 TWh) was composed of nuclear (42%), renewables and biofuels (33%), natural gas (18%), and other sources (8%). Belgium's gas system has a peak capacity of 57 GW, far exceeding the electricity system's 14 GW, making gas infrastructure an indispensable source of seasonal and daily flexibility, especially as intermittent renewables expand.

› 7.3.3 Ten-Year Network Development Plan and Investments

Fluxys Belgium's 2025–2034 investment plan amounts to an indicative €8.3 billion. Over 90% of this plan is directed towards infrastructure for low-carbon molecules and reducing the company's own greenhouse gas emissions. The main investment categories are: hydrogen infrastructure (the largest share), CO₂ infrastructure, facilitating green gas integration into the natural gas network, reducing Fluxys's own GHG emissions, and other investments.

Hydrogen and CO₂ transmission in Belgium are managed through three separate Fluxys Belgium affiliates: *Fluxys Hydrogen* (100% owned, nominated as the Belgian Hydrogen Network Operator for 20 years), *Fluxys c-grid* (77.5% owned, CO₂ network operator in Wallonia and Flanders), and *Fluxys c-grid Antwerp* (70% owned, CO₂ network operator application submitted for the Antwerp port cluster).

› 7.3.4 Biomethane

Biomethane is highlighted as a “no-brainer” near-term decarbonisation lever. Produced from organic waste, livestock effluents, and agriculture-related feedstocks through anaerobic digestion and purification, biomethane is chemically identical to natural gas and can be injected directly into the existing grid. The EU27+UK biomethane supply potential is estimated at 1,110 TWh/year by 2050, covering applications in transport (566 TWh), power (300 TWh), buildings (178 TWh), and industry (66 TWh).

7.4 Hydrogen Infrastructure

› 7.4.1 Hydrogen Colours and Production Pathways

Hydrogen is produced through several routes with varying carbon intensities:

- **Grey hydrogen:** Steam methane reforming (SMR) without carbon capture.
- **Blue hydrogen:** SMR or autothermal reforming with carbon capture – classified as Low Carbon H₂.
- **Green hydrogen (RFNBO):** Electrolysis powered by renewable electricity (wind or solar).
- **Pink hydrogen:** Electrolysis powered by nuclear energy – Low Carbon H₂.
- **Turquoise hydrogen:** Pyrolysis of natural gas into hydrogen and solid carbon – Low Carbon H₂.
- **White hydrogen:** Naturally occurring geological deposits.

Compared to natural gas (CH₄), hydrogen (H₂) has a much wider flammability range (4–75% vs. 5–15%), a lower minimum ignition energy (0.017 mJ vs. 0.3 mJ), a lower volumetric energy density (3.54 kWh/m³ vs. 11.3 kWh/m³), but a higher gravimetric energy density (39.4 kWh/kg vs. 13.3 kWh/kg). Its liquefaction temperature is –253°C, compared to –162°C for LNG, making liquid hydrogen shipping technically challenging and energy-intensive (33% liquefaction losses vs. 10% for LNG).

› 7.4.2 Belgian Hydrogen Backbone

Fluxys Hydrogen is developing a phased hydrogen backbone connecting Belgium’s industrial valleys (Ghent, Antwerp, Brussels, Charleroi, Liège, Mons) and neighbouring countries (Netherlands, Germany, France, Luxembourg). Key targets include:

- 30 TWh/year transmission capacity by 2030;
- First infrastructure operational in 2026 (Zelzate–Kallo and Antwerp South sections, Final Investment Decision already taken);
- Cross-border connections with OGE (Germany), Creos (Luxembourg), natTran (France), and HyNetwork Services (Netherlands);
- 1,200 km of H₂ pipelines by 2050.

At the European level, the European Hydrogen Backbone vision – developed by 33 infrastructure operators – targets 31,000 km of H₂ infrastructure by 2030 and 53,000 km by 2040, fed by five supply corridors (North Sea, Nordic and Baltic regions, East and Southeastern Europe, North Africa, and Southwest Europe). The EU27 hydrogen demand is projected at 1,600–2,500 TWh/year by 2050, primarily for industry (700–800 TWh), transport (500–700 TWh), power (200–500 TWh), and buildings (200–500 TWh).

Import of low-carbon hydrogen from outside the EU is expected to be necessary, using carriers such as liquefied hydrogen, ammonia, methanol, LOHC (Liquid Organic Hydrogen Carriers), or synthetic methane. The Belgian Hydrogen Import Coalition (including Fluxys, ENGIE, DEME, EXMAR, and Port of Antwerp-Bruges) has validated the technical and economic feasibility of green hydrogen imports from regions such as Australia, Chile, Oman, Morocco, and Iberia.

Fluxys is also developing an open-access ammonia import terminal (Advario project at Antwerp, targeted operational by 2030), with NH₃ cracking into H₂ and connection to the Fluxys H₂ network. Additionally, the BE-HyStore project, in collaboration with Ghent University, is exploring underground hydrogen storage at Loenhout with a target capacity of 2.4 TWh.

7.5 CO₂ Infrastructure

› 7.5.1 CO₂ Transport and the CCUS Value Chain

Carbon capture and storage/utilisation (CCS/CCU) is increasingly recognised as a necessary mitigation pathway in the EU Industrial Carbon Management Strategy. The EU27 target for CO₂ captured for storage and utilisation is 450 Mt/year by 2050 (from approximately 50 Mt in 2030). CO₂ is transported in dense phase (supercritical, above 74 bara and 31°C) to maximise pipeline efficiency.

Fluxys's CO₂ value chain covers onshore transmission by pipeline, terminalling (liquefaction, buffer storage, ship loading/unloading, train unloading), and marine shipping to offshore storage or sequestration sites. Sources include industrial carbon capture facilities; destinations include geological storage sites in the North Sea (Norway, UK) and the Netherlands.

› 7.5.2 Belgian CO₂ Network Development

The Fluxys c-grid system for Belgium envisions a national CO₂ backbone connecting industrial clusters across Wallonia and Flanders, with ambitions to offer 30 MTPA (million tonnes per annum) of transmission capacity by 2030, with first infrastructure ready by end-2028. The network includes connections towards France (15–30 Mt/year), Germany and the East (34–73 Mt/year), and the North Sea (9–16 Mt/year). Key export projects include:

- **Zeebrugge offshore CO₂ pipeline** (joint venture with Equinor): 27 MTPA capacity, commissioning target 2031, for sequestration in Norway and/or the UK.
- **Dunkirk Marine CO₂ Export Terminal** (joint venture with Air Liquide and Dunkerque LNG): 1.5 MTPA initial capacity, commissioning target 2028, co-funded by the EU.
- **Antwerp@C Marine CO₂ Export Terminal** (joint venture with Air Liquide and Port of Antwerp-Bruges): up to 10 MTPA, commissioning target 2029, co-funded by the EU.

Within the Antwerp cluster, a Final Investment Decision has already been taken for the first pipeline section (Lillo–Kallo–Zandvliet axis), which connects the port area to both Rotterdam (Aramis project in the Netherlands) and Zeebrugge. Construction is partly co-funded by the EU's Connecting Europe Facility (CEF).

› 7.5.3 Tariff Structure and Derisking for CO₂ Infrastructure

The development of CO₂ infrastructure faces a chicken-and-egg challenge: the market is nascent, bookings are well below the reference volume (Qref) used in regulated tariff formulas. A high standalone tariff in early years would be unacceptable to first-mover industrial emitters. The proposed solution involves setting an initial tariff aligned with the long-term target tariff, with the resulting revenue gap compensated by government support (e.g., in the form of conditional loans repayable as the market scales up). As capacity bookings grow towards the full Qref, excess revenues cover the repayment. This model mirrors a public-investment logic for new infrastructure with high social value.

7.6 The North Sea Integration Model

Fluxys, in partnership with the University of Liège and funded by Belgium's Energy Transition Fund (SPF Économie), has developed a multi-energy system simulation model covering all 10 North Sea countries (Belgium, France, Germany, Netherlands, Luxembourg, Denmark, Norway, Ireland, the United Kingdom, and Austria). The model optimises on an hourly basis the interactions between electricity, methane (CH₄), hydrogen (H₂ and derivatives), and CO₂, targeting carbon neutrality by 2050 at minimum system cost while ensuring energy demand is met at all times.

Key modelling characteristics include linear optimisation over investments and dispatch, perfect foresight (no uncertainty), and no intra-country distance assumption. Inputs include final energy demand (hourly), renewable production potentials (hourly), import assumptions, cost parameters (CAPEX, OPEX, WACC), technical parameters, and CO₂ reduction targets. Outputs include optimal infrastructure investments, hourly dispatch flows, global system cost, marginal prices, energy not served, and CO₂ emissions by country.

Key findings from the North Sea scenario for 2050 include: renewables (solar, wind, hydro) cover electricity demand and green hydrogen production; Norwegian pipeline gas, LNG, and bio-methane cover remaining gas demand and low-carbon hydrogen production via SMR with carbon capture; gas-fired power plants (CCGT and OCGT) run at full capacity only 3% of the year and supply approximately 1% of total electricity, but remain

indispensable for security of supply during periods of low renewable generation; and CO₂ capture is split equally between biological credits (biomass and bio-CH₄) and geological capture from low-carbon hydrogen production and industrial processes. An interactive demonstrator of the Belgian model results is publicly accessible at <https://integrationdemonstrator.github.io/>.

» Chapter 8 — Carbon capture

8.1 Introduction: Context and Motivation

Atmospheric CO₂ concentrations have now permanently exceeded 400 ppm, accompanied by a measurable rise in global temperatures. This situation calls for a combination of mitigation strategies, including reductions in energy consumption, substitution of fossil fuels with natural gas or biomass, deployment of renewable and nuclear energy, and Carbon Capture, Utilization, and Storage (CCUS). Among these, CCUS stands out as one of the few approaches capable of addressing emissions from hard-to-decarbonize industrial sectors at a large scale.

Industrial flue gases represent the primary target for capture technologies. Depending on the source, the CO₂ molar fraction (y_{CO_2}) ranges from approximately 3% in combined-cycle natural gas plants (CCGT) to around 15% in coal-fired power stations. Lime and cement plants are particularly CO₂-intensive due to the calcination reaction ($\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$), which can drive y_{CO_2} up to 20–30%, making them prime candidates for capture retrofits.

A complementary approach is **Direct Air Capture (DAC)**, which extracts CO₂ directly from ambient air (at ≈ 400 ppm). While energetically more demanding—requiring 19–21 kJ/mol CO₂ compared to 1–9 kJ/mol for industrial flue gas sources—DAC offers a key land-use advantage: it requires 500 to 1500 times less surface area than forests to capture an equivalent quantity of CO₂. A well-known example is the ORCA facility in Iceland, capturing 4000 t_{CO₂}/year using renewable geothermal energy. In the CCUS value chain, captured CO₂ is transported by pipeline or ship and either stored geologically (with an estimated worldwide capacity of 2000–20 000 Gt_{CO₂}) or utilized in a growing range of industrial, chemical, and food applications, with a utilization potential estimated at 1–7 Gt_{CO₂}/year.

The **EU Emissions Trading Scheme (ETS)** provides an economic incentive: with CO₂ allowance prices currently above 70 €/t_{CO₂}, large emitters face a financial pressure that makes investment in capture technologies increasingly competitive.

8.2 CO₂ Capture Strategies

Three broad combustion-side strategies can be distinguished:

- **Pre-combustion capture:** The fuel (natural gas, coal, or biomass) is first reformed or gasified, then subjected to a water-gas shift reaction, producing a high-pressure H₂/CO₂ mixture. The CO₂ is separated before combustion, and the purified hydrogen is subsequently burned.
- **Oxyfuel combustion:** The fuel is burned in a nearly pure oxygen atmosphere, yielding a flue gas composed predominantly of CO₂ and water (>80% CO₂). Water is condensed, leaving a concentrated CO₂ stream ready for compression and storage with minimal further treatment.
- **Post-combustion capture:** CO₂ is removed from the flue gas after a conventional combustion process. This so-called “end-of-pipe” approach is the most technologically mature and the most easily retrofittable to existing industrial facilities, as it does not require modification of the combustion unit itself.

For each of these strategies, several physico-chemical separation techniques exist: gas-liquid absorption, gas-solid adsorption, membrane separation (gas-gas or gas-liquid), cryogenic condensation, calcium looping, chemical looping combustion, and biological approaches using micro-algae.

8.3 Gas-Liquid Absorption: The Reference Technology

› 8.3.1 Process Description

Gas-liquid (G-L) absorption remains the most industrially proven technology for post-combustion CO₂ capture. The process, originally patented by R. R. Bottoms in 1930 for acid gas separation, relies on two coupled columns:

- An **absorber**, operating at approximately 40°C and 1.2 bar, where the flue gas contacts a descending lean solvent and CO₂ is chemically or physically absorbed.
- A **regenerator (stripper)**, operating at around 120°C and 2 bar, where heat is applied to release concentrated CO₂ and regenerate the solvent for recirculation.

The **regeneration energy** has three contributions:

$$E_{\text{regen}} = \Delta H_{\text{Reaction}} + \Delta H_{\text{Sensible}} + \Delta H_{\text{VAP, solvent}}$$

where $\Delta H_{\text{Reaction}}$ is the energy needed to break the solvent–CO₂ bond, $\Delta H_{\text{Sensible}}$ is the energy required to heat the solvent to the stripping temperature, and $\Delta H_{\text{VAP, solvent}}$ accounts for solvent vaporization. For a reference 30% monoethanolamine (MEA) solution, this total is approximately 3.7 GJ/t_{CO₂}.

› 8.3.2 Solvent Generations

Solvent selection is governed by multiple interacting criteria: reaction kinetics with CO₂, physico-chemical properties (density, viscosity, diffusivity, solubility), regeneration energy, solvent degradation behavior with CO₂, oxygen, SO_x and NO_x, industrial availability, and cost.

First-generation solvents are single-component aqueous solutions, typically at 30% concentration:

- *Primary amines* such as MEA react rapidly via a zwitterion intermediate to form carbamates, offering high reactivity but high regeneration energy and significant degradation susceptibility.
- *Tertiary amines* such as MDEA (N-methyldiethanolamine) react via bicarbonate formation; they are slower but require less regeneration energy.
- *Sterically hindered amines* (SHA) such as AMP (2-amino-2-methyl-1-propanol) form very unstable carbamates that readily hydrolyze to bicarbonate, combining higher loading capacity with lower regeneration energy.
- *Cyclic diamines* such as piperazine (PZ) act as absorption activators due to their fast kinetics, often used as promoters in blends.

Second-generation solvents are blends of two or more first-generation solvents, exploiting complementary properties. The activation mechanism—for example, MDEA activated by PZ (commercialized by BASF as aMDEA in the early 1980s) or the more recent CESAR-1 formulation (AMP + PZ)—significantly reduces regeneration energy while maintaining adequate absorption rates. Hybrid solvents combining chemical and physical absorbents (e.g., Sulfinol) are also in this category.

Third-generation solvents include ionic liquids (ILs), which are molten salts liquid at ambient conditions with very low vapor pressure and high thermal stability (e.g., imidazolium, pyridinium, ammonium families), and amino acid salts such as potassium proline (KPro) or potassium sarcosinate (Ksar), which replace a hydrogen atom of classical amines by a cation, yielding non-volatile, potentially biodegradable solvents.

Fourth-generation solvents represent the current research frontier and include:

- *Enzymatic solvents*: A conventional solvent (amine or K₂CO₃) is combined with the enzyme carbonic anhydrase, which catalyzes the CO₂ hydration reaction and dramatically accelerates the absorption rate.
- *Demixing (phase-change) solvents*: Upon CO₂ absorption, the loaded solvent splits into two liquid phases. Only the CO₂-rich phase is sent to the regenerator, substantially reducing the volume of liquid requiring heating and thus lowering the regeneration energy by 20–30% compared to MEA. The DMX[®] process developed by IFP Energies Nouvelles and the DEEA-MAPA system are leading examples.

A comparison of specific reboiler duties across solvent generations illustrates the progress achieved: from 3.7–4.5 GJ/t_{CO₂} for MEA (circa 1990–2005) down to approximately 2.2 GJ/t_{CO₂} for the best phase-change solvent systems demonstrated around 2015.

› 8.3.3 Process Improvements and Degradation

Beyond solvent selection, process configuration plays a major role. Configurations studied include intercooling of the absorber, rich-solvent vapor recompression (RVC), lean vapor compression (LVC), split-flow arrangements, staged stripper feeding, partial flash regeneration, and multi-column schemes. Simulation results show that combining MDEA/PZ with rich vapor compression and intercooling can reduce E_{regen} by up to **35%** relative to the MEA base case, reaching a minimum of approximately 2.19 GJ/t_{CO₂}.

Packing selection also influences performance: structured packings (Flexipac, Mellapak, BX) consistently outperform random packings such as Raschig or Pall rings in terms of reboiler duty and solvent flow rate.

Solvent degradation is a major operational challenge. Mechanisms include carbamate polymerization, oxazolidinone and imidazolidinone formation via reaction with CO₂, thermal degradation at regeneration temperatures, and oxidative degradation by O₂, SO_x, and NO_x present in flue gas. Degradation products cause corrosion of carbon steel equipment and foaming in the absorber column, both of which increase operating costs and maintenance requirements.

8.4 Other Capture Technologies

› 8.4.1 Gas-Solid Adsorption

Adsorption relies on the physical attraction between CO₂ molecules and active sites at a solid adsorbent surface. Regeneration is achieved by either **Temperature Swing Adsorption (TSA)**—heating the bed to desorb CO₂—or **(Vacuum) Pressure Swing Adsorption ((V)PSA)**—reducing total pressure to lower the CO₂ partial pressure and release it. Common adsorbents include activated carbons (derived from coconut, coal, or petroleum coke), zeolites (microporous alkali silicoaluminates), and Metal-Organic Frameworks (MOFs), which offer highly tunable pore structures and selectivities. These materials are currently at TRL 6 and represent an active research front.

› 8.4.2 Membranes

Two membrane configurations are relevant to CO₂ capture:

Gas-gas membranes act as semi-permeable barriers exploiting differences in molecular permeability. The driving force is the pressure difference across the membrane; CO₂ preferentially permeates to the low-pressure side. Membrane types range from cheap polymeric membranes (limited by thermal stability above 400°C) to more robust but expensive zeolite, palladium, or ceramic membranes. Their main advantages are compactness (no regeneration step required, smaller footprint than packed columns) and low maintenance. However, they are most competitive for small-to-medium gas flow rates and/or high CO₂ concentrations; at large industrial scales and low concentrations, amine absorption remains more cost-effective. Hybrid processes combining membranes and amines are particularly attractive at intermediate scales.

Gas-liquid membranes combine membrane and solvent technologies: a microporous membrane provides the gas-liquid interface without mixing the phases, while selectivity is governed entirely by the choice of absorption liquid on the permeate side. The membrane's sole role is to maximize interfacial area.

› 8.4.3 Cryogenic Processes

CO₂ can be separated by cooling the flue gas until it condenses as a liquid. Often preceded by a PSA step to enrich the CO₂ stream, as in Air Liquide's CRYOCAP™ FG process, this approach is best suited to streams already rich in CO₂ (>60%) and at elevated pressure, such as in ammonia production. Its main disadvantages are a high refrigeration energy demand and the need to remove water and other condensable impurities upstream. Its chief advantage is the production of a highly pure liquid CO₂ ready for pipeline transport.

› 8.4.4 Calcium Looping and Chemical Looping Combustion

Calcium Looping (CaL) uses CaO as a regenerable solid sorbent. At ≈650°C in a carbonator, CaO reacts with CO₂ to form CaCO₃. The loaded sorbent is then transferred to a calciner operating at ≈950°C, where it is regenerated and releases a concentrated CO₂ stream. The purged CaO can feed a cement kiln, offering synergies with the cement industry. A related technology developed by LEILAC uses indirect heated calcination to directly capture process CO₂ from cement and lime production without additional chemicals.

Chemical Looping Combustion (CLC) is an alternative to oxyfuel combustion in which a metal oxide (e.g., Fe₂O₃, NiO, CuO, Mn₃O₄) circulates between two fluidized-bed reactors. In the fuel reactor, the oxide is

reduced by the fuel, producing CO_2 and H_2O ; water is easily condensed to yield pure CO_2 . In the air reactor, the reduced metal is re-oxidized, releasing nitrogen-depleted air. This inherently separates CO_2 without the energy cost of an air separation unit.

› 8.4.5 Biological Approaches: Micro-algae

Micro-algae consume CO_2 and light via photosynthesis in photobioreactors (open or closed). After separation from the culture medium, the algal biomass can be processed into biofuels or used in an ethanol production cycle. The main limitation remains the large land area required and the energy demand for mixing and harvesting, though algae can simultaneously remove other pollutants such as NO_x .

8.5 Industrial Deployment and Outlook

As of 2025–2026, the global CCUS project pipeline has grown substantially: the Global CCS Institute reports 734 facilities in total (a 17% year-on-year increase), with 77 in operation, 47 under construction, and a combined capture capacity of 513 Mt_{CO_2} /year across all development stages—more than tripled since 2017.

Notable industrial references include:

- **Boundary Dam (Canada):** The world's first commercial post-combustion CCS unit on a coal-fired power plant (SaskPower, Unit 3), capturing up to 1 Mt_{CO_2} /year and simultaneously removing 100% of SO_2 emissions. Operational data confirmed a 90% CO_2 capture rate, equivalent to removing approximately 250 000 cars from the road annually.
- **Niederaussem (Germany):** A pilot plant attached to a 965 MW lignite power plant, treating $\approx 1550 \text{ Nm}^3/\text{h}$ of flue gas at 15% CO_2 , achieving $\approx 90\%$ capture efficiency. The pilot has validated MEA, BASF's OASE[®] blue (20% lower specific energy than MEA), and CESAR-1 (AMP + PZ) at packing heights of 16 m.
- **Technology Centre Mongstad (TCM), Norway:** The world's largest CO_2 capture test center, providing open-access infrastructure for testing both amine and chilled ammonia processes on real flue gases from a combined heat and power plant and a refinery.
- **Norcem Brevik Cement Plant (Norway):** A 400 000 t/year capture demonstration project exploiting waste heat from the kiln for solvent regeneration (50% capture rate), now fully constructed—a first for the cement sector.
- **Duiven Waste Incineration Plant (Netherlands):** 165 t_{CO_2} /day captured by amine absorption, illustrating scalability to waste-to-energy facilities.

Despite these advances, the primary remaining barrier to widespread deployment is economic rather than technical. Capture costs must decrease further to align with carbon prices and avoid excessive energy penalties. The pathway forward involves developing advanced solvents with lower regeneration energies, optimizing process configurations, scaling up novel capture concepts (membranes, adsorption, CaL), and building the CO_2 transport and storage infrastructure required to connect emitters with geological reservoirs.

As a closing note, CCUS is not a substitute for emissions reduction at source. Renewable energy deployment, energy efficiency improvements, and demand reduction remain essential. CCUS provides a complementary, necessary tool for sectors where direct decarbonization is technically or economically constrained—and for removing historical CO_2 already in the atmosphere through BECCS or DAC. In the words of the author: *“The best captured CO_2 is the one that we have not emitted.”*

» Chapter 9 – Elia

» Chapter 10 – Nuclear

» Chapter 11 – Aluminium

» Chapter 12 – Energy in Africa

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» Chapter 14 – Video and reflexive process